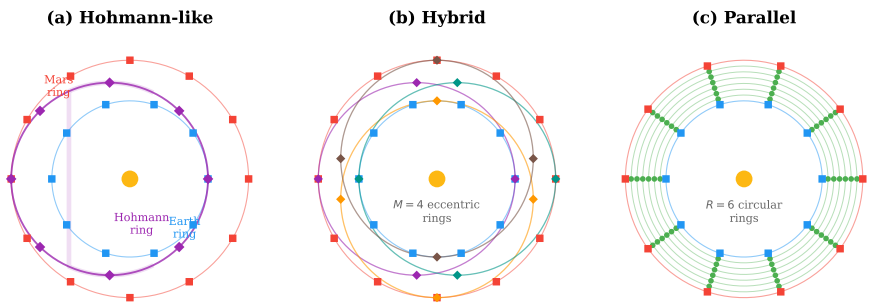

Figure 1: Schematic comparison of the three relay topologies (top-down heliocentric view). **(a)** Hohmann-like: one eccentric transfer ring connects the Earth and Mars circular rings at perihelion and aphelion. **(b)** Hybrid: M eccentric rings with staggered arguments of perihelion improve coverage. **(c)** Parallel: R concentric circular rings create N independent radial routes.

1 Topology Comparison: Hohmann-Like, Hybrid, and Parallel

Having established in **??** that a constellation of many inexpensive relay satellites is the preferred architecture, the next design decision is how to *arrange* those satellites. The topology determines how data flows from Earth to Mars: how many independent paths exist, which link is the bottleneck, and how the constellation responds to individual satellite failures. We evaluate three candidate topologies—Hohmann-like, hybrid, and parallel—against four criteria: aggregate throughput scaling, redundancy, failure tolerance, and hardware complexity per satellite. Figure 1 illustrates the three topologies schematically.

1.1 Hohmann-Like (Three-Ring Series)

The most intuitive relay topology uses three orbital rings. A circular *Earth ring* of N_E satellites orbits at heliocentric radius r_{E}. A circular *Mars ring* of N_M satellites orbits at r_{M}. Between them, a single *Hohmann ring* of N_H satellites follows an elliptical transfer orbit whose perihelion coincides with r_{E} and whose aphelion coincides with r_{M}. Data flows in series: ground stations hand off to the Earth ring, the Earth ring relays to the Hohmann ring near perihelion, the Hohmann ring relays to the Mars ring near aphelion, and the Mars ring delivers to the surface.

Each satellite on the Earth and Mars rings carries two laser terminals (circumferential, to maintain the ring) plus one inter-ring terminal pointing toward the Hohmann ring. Each Hohmann-ring satellite carries two circumferential terminals plus two inter-ring terminals (one Earth-facing, one Mars-facing), for a total of four.

The bottleneck is the inter-ring link. At perihelion, Hohmann-ring satellites

are co-orbital with the Earth ring, and the link distance is set by the angular separation between the nearest satellites on each ring—of order $\pi r_{\mathrm{E}}/N_E$ for the Earth ring. At aphelion, the same applies at Mars: the link distance is of order $\pi r_{\mathrm{M}}/N_M$. Between these extremes, Hohmann-ring satellites spend the majority of their orbital period at intermediate heliocentric distances where they are far from both planetary rings. During cruise, the link to the nearest planetary ring spans a distance of order $D_{\mathrm{EM}}/2 \approx 39$ Mkm, and the inverse-square law reduces per-link capacity to

$$C_{\mathrm{cruise}} \sim \frac{K}{(D_{\mathrm{EM}}/2)^2} = \frac{4K}{D_{\mathrm{EM}}^2} \tag{1}$$

For the baseline parameters ($K = 9 \times 10^8\,\mathrm{Gbps} \cdot \mathrm{km}^2$, $D_{\mathrm{EM}} = 78 \times 10^6\,\mathrm{km}$), this gives approximately 0.6 kbps—far below any useful threshold. The Hohmann ring is only effective when its satellites are near perihelion or aphelion; the rest of the orbit is a dead zone.

Because the interplanetary segment passes through a single Hohmann ring, the topology has only one data path between Earth and Mars at any given time. This creates a single point of failure: the loss of the active Hohmann-ring relay severs the link entirely. The Earth and Mars rings provide local redundancy (a failed satellite on either circular ring can be bypassed via its neighbours), but this is irrelevant if the interplanetary hop is broken.

Throughput does not scale well with constellation size. Adding satellites to the Earth or Mars rings improves local connectivity but does nothing for the interplanetary bottleneck. Adding satellites to the Hohmann ring increases the number of potential relay nodes near perihelion and aphelion, but only one is active at a time (the one closest to the handoff point), so throughput remains limited to a single inter-ring link.

1.2 Hybrid (Multiple Eccentric Rings)

The hybrid topology retains the Earth and Mars circular rings of the Hohmann-like configuration but replaces the single Hohmann ring with M eccentric rings, each on its own transfer orbit. All M orbits share the same semi-major axis and eccentricity (perihelion at r_{E}, aphelion at r_{M}) but have their arguments of perihelion equally spaced at intervals of $360°/M$. This angular staggering ensures that, at any instant, the M eccentric rings are distributed around the Sun at different phases of their orbits, so that at least some satellites are always near one of the two planetary rings.

The motivation is to address the dead-zone problem of the Hohmann-like topology. With M eccentric rings, the probability that *some* eccentric satellite is near perihelion or aphelion at any given time increases with M. For large enough M, the constellation can maintain near-continuous handoff between the Earth ring and whichever eccentric ring is currently closest, and likewise between the Mars ring and its nearest eccentric neighbour.

However, the fundamental throughput bottleneck remains. Each eccentric ring still traverses the full Earth–Mars gap, and a satellite on any such ring

spends the majority of its orbital period at intermediate heliocentric distances where the link to either planetary ring spans tens of millions of kilometres. The inverse-square penalty during cruise (eq. (1)) applies identically to every eccentric ring. Staggering the arguments of perihelion improves *availability*—the fraction of time that at least one usable relay exists—but does not improve the *throughput* of any individual relay during cruise.

The hybrid topology does gain a measure of redundancy over the Hohmann-like configuration. With M eccentric rings, the loss of one ring still leaves $M - 1$ alternative interplanetary paths. But each path independently suffers the same cruise-phase throughput floor, so the aggregate capacity scales at best linearly with M—and only during the favourable orbital phases when an eccentric satellite is near a planetary ring. During cruise, additional eccentric rings add satellites that contribute negligible throughput.

Hardware complexity also increases. Each eccentric-ring satellite requires four laser terminals (two circumferential, one Earth-facing, one Mars-facing), and the pointing geometry changes continuously along the eccentric orbit. Satellites on the Earth and Mars rings need additional terminals to maintain links with whichever eccentric ring is currently nearby. The total terminal count per satellite is 3–4, compared with 2 for the parallel topology.

Figure 2 quantifies the dead-zone problem. For a single Hohmann ring ($M = 1$), the best available inter-ring link capacity drops by over eight orders of magnitude between the perihelion/aphelion peaks and the mid-orbit trough. Adding eccentric rings ($M = 3, 6$) raises the envelope by introducing peaks at intermediate phases, but the capacity between peaks remains far below the constant throughput achievable with the parallel topology.

1.3 Parallel (Concentric Circular Rings)

The parallel topology distributes the S satellites across R concentric heliocentric circular orbits ("rings") between Earth and Mars. Each ring contains $N = S/R$ satellites evenly spaced around its circumference. Every satellite carries exactly two laser terminals: one pointing radially inward to the nearest satellite on the adjacent inner ring, and one pointing radially outward to the nearest satellite on the adjacent outer ring. There are no circumferential (intra-ring) links.

This arrangement creates N independent parallel routes from Earth to Mars, each consisting of a radial chain of R inter-ring hops. The throughput of each route is limited by its bottleneck link (the outermost inter-ring gap, as derived in section 2.3), and the total constellation throughput is the sum over all N routes:

$$T_{\text{parallel}} \;=\; N \cdot \frac{K}{d_{\text{max}}^2} \tag{2}$$

As shown in section 2.5, optimising the distribution between rings and routes yields $T_{\text{opt}} \propto S^{3/2}$. This super-linear scaling is the parallel topology's defining advantage: every satellite added improves both the per-route capacity (by shortening inter-ring distances) and the aggregate capacity (by adding routes).

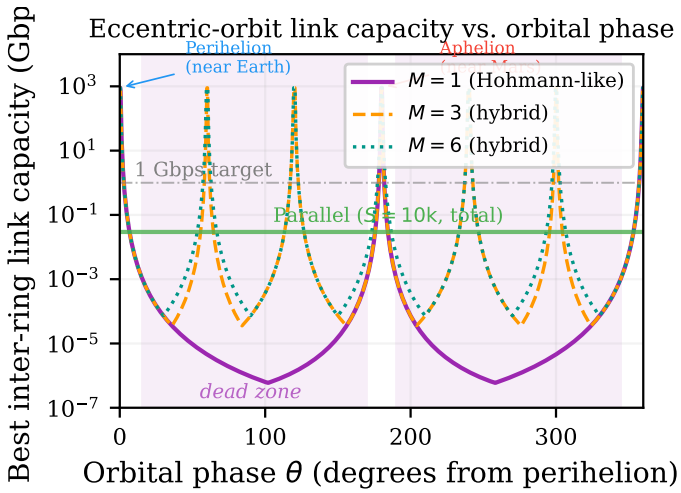

Figure 2: Best available inter-ring link capacity as a function of orbital phase for the Hohmann-like ($M = 1$) and hybrid ($M = 3, 6$) topologies. The horizontal green line shows the total throughput of a parallel constellation with $S = 10{,}000$ satellites at optimal ring allocation. Baseline parameters: $f = 1$, $K = 9 \times 10^8$ Gbps $\cdot$ km^2.

Redundancy is intrinsic. The loss of a single satellite removes one route from one ring but leaves all other $N - 1$ routes intact. Throughput degrades gracefully—by a fraction $1/N$—rather than catastrophically. For a constellation with $N = 100$ routes per ring, losing one satellite reduces total throughput by 1%.

The hardware requirement is minimal: two laser terminals per satellite, the fewest of any topology considered. The two terminals point radially (inward and outward), and each satellite participates in exactly one route. This radial-only pointing simplifies acquisition and tracking, since adjacent rings are separated by a fixed, predictable gap.

Scalability follows naturally. Adding satellites to existing rings increases N and therefore the number of parallel routes. Adding new rings between existing ones increases R, shortening inter-ring distances and boosting per-route capacity. Both operations improve total throughput without requiring any change to the satellite hardware or link architecture.

1.4 Topology Comparison

Table 1 summarises the key properties of the three topologies. The Hohmann-like and hybrid configurations both suffer from the same fundamental limitation: their interplanetary segments traverse the full Earth–Mars gap on eccentric orbits, spending most of each orbital period at distances where the inverse-square law renders throughput negligible. The parallel topology avoids this

Table 1: Comparison of relay constellation topologies for S total satellites.

Property	**Hohmann-like**	**Hybrid**	**Parallel**
Interplanetary segment	1 eccentric ring	M eccentric rings	R circular rings
Aggregate throughput scaling	Single-link limited	$\sim M$ (favourable phase)	$S^{3/2}$ (N routes
Parallel data paths	1	$\leq M$	$N = S/R$
Redundancy	Local rings only	Local + partial interplanetary	Per-route
Single-satellite failure	Interplanetary outage	$1/M$ path loss	$1/N$ throughput l
Laser terminals / satellite	3–4	3–4	2
Cruise dead zone	Yes (severe)	Yes (per ring)	None
Scalability	Add to rings	Add eccentric rings	Add to rings or add

entirely by breaking the interplanetary gap into many short radial hops on circular orbits, achieving both higher throughput and intrinsic redundancy with simpler hardware.

We adopt the parallel concentric-ring topology for the remainder of this paper. Section 2 develops the full throughput model, derives the optimal allocation between rings and routes, and establishes the $S^{3/2}$ scaling law that governs the constellation's performance.

2 Throughput Analysis

2.1 Link Budget Model

The capacity of a single free-space optical link at distance d is modelled by the inverse-square law:

$$C(d) \;=\; f\, C_0 \left(\frac{d_0}{d}\right)^2 \tag{3}$$

where $C_0 = 100\,\text{Gbps}$ is the baseline capacity at reference distance $d_0 = 3{,}000\,\text{km}$ (consistent with NASA TBIRD-class demonstrations), and f is a technology improvement factor reflecting advances in laser power, receiver sensitivity, and modulation efficiency. We define the constant

$$K \;=\; f\, C_0\, d_0^2 \tag{4}$$

With $f = 1$ (no improvement), $K = 9 \times 10^8\ \text{Gbps} \cdot \text{km}^2$.

2.2 Constellation Geometry

Consider R relay rings placed on concentric heliocentric circular orbits between Earth's orbit (radius $r_\text{E} \approx 150$ Mkm) and Mars' orbit (radius $r_\text{M} \approx 228$ Mkm average, 249 Mkm at apoapsis). Each ring contains N satellites evenly spaced

around its circumference. The total relay satellite count is $S = N \times R$. Each satellite carries exactly two laser terminals — one pointing radially inward to the adjacent inner ring, one pointing radially outward to the adjacent outer ring. There are no circumferential (intra-ring) links. This creates N parallel *routes*, each a radial chain hopping from ring 1 (nearest Earth) through to ring R (nearest Mars).

Ring i ($i = 1, \ldots, R$) sits at heliocentric radius

$$r_i \;=\; r_{\mathrm{E}} + i\,\frac{D_{\mathrm{EM}}}{R+1} \tag{5}$$

where $D_{\mathrm{EM}} = r_{\mathrm{M}} - r_{\mathrm{E}}$. Using average orbital radii, $D_{\mathrm{EM}} \approx 78$ Mkm; at Mars apoapsis, $D_{\mathrm{EM}} \approx 99$ Mkm. The inter-ring radial gap is

$$d_{\mathrm{radial}} \;=\; \frac{D_{\mathrm{EM}}}{R+1} \tag{6}$$

and the arc spacing between adjacent satellites on ring i is $2\pi r_i/N$.

2.3 Worst-Case Inter-Ring Link Distance

A satellite on ring i connects to the *nearest* satellite on ring $i+1$. In the worst case, the nearest neighbor is offset by half the arc spacing on ring $i+1$. The worst-case link distance between adjacent rings is therefore

$$d_{\mathrm{worst}} \;=\; \sqrt{d_{\mathrm{radial}}^2 + \left(\frac{\pi\, r_{i+1}}{N}\right)^2} \tag{7}$$

Since ring circumference increases with heliocentric radius, the outermost inter-ring gap (ring R to Mars) always has the largest angular offset and therefore the lowest throughput per link. This outermost gap is the *bottleneck*. Using $r_{i+1} \approx r_{\mathrm{M}}$ for the worst case (fig. 3):

$$d_{\max} \;=\; \sqrt{\left(\frac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\frac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{8}$$

The throughput of a single route (limited by the bottleneck link) is

$$C_{\mathrm{route}} \;=\; \frac{K}{d_{\max}^2} \;=\; \frac{K}{\left(\dfrac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\dfrac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{9}$$

and the total constellation throughput, summed over all N parallel routes, is

$$T(N,R) \;=\; N \cdot C_{\mathrm{route}} \;=\; \frac{N\,K}{\left(\dfrac{D_{\mathrm{EM}}}{R+1}\right)^2 + \left(\dfrac{\pi\, r_{\mathrm{M}}}{N}\right)^2} \tag{10}$$

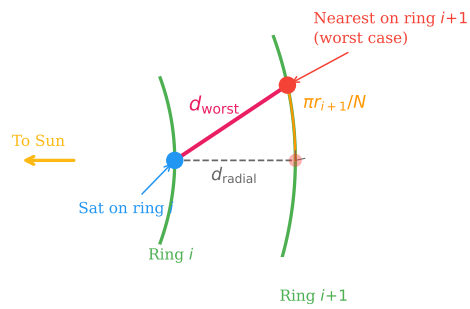

Figure 3: Worst-case link geometry between adjacent rings. The link distance d_{worst} is the hypotenuse of the radial gap d_{radial} and the angular half-gap $\pi r_{i+1}/N$.

2.4 Optimal Distribution: Ring Count vs. Route Count

For a fixed total satellite count $S = NR$, we substitute $N = S/R$ into eq. (10). Approximating $R + 1 \approx R$ for large R:

$$T(S, R) \;=\; \frac{S\,K}{\dfrac{D_{\text{EM}}^2}{R} + \dfrac{\pi^2 r_{\text{M}}^2\,R^3}{S^2}} \tag{11}$$

Maximising T is equivalent to minimising the denominator

$$g(R) \;=\; \frac{D_{\text{EM}}^2}{R} + \frac{\pi^2 r_{\text{M}}^2\,R^3}{S^2} \tag{12}$$

Setting $g'(R) = 0$:

$$g'(R) = -\frac{D_{\text{EM}}^2}{R^2} + \frac{3\,\pi^2 r_{\text{M}}^2\,R^2}{S^2} = 0$$

$$R^4 = \frac{D_{\text{EM}}^2\,S^2}{3\,\pi^2\,r_{\text{M}}^2}$$

$$R_{\text{opt}} = \sqrt{\frac{D_{\text{EM}} \cdot S}{\sqrt{3}\,\pi\,r_{\text{M}}}} \tag{13}$$

or, in words:

> *The optimal number of relay rings equals the square root of the Earth–Mars distance times the total satellite count, divided by $\sqrt{3}\,\pi$ times Mars' orbital radius.*

Numerically, with $D_{\text{EM}} = 78$ Mkm (average) and $r_{\text{M}} = 228$ Mkm:

$$R_{\text{opt}} \approx 0.00251\,\sqrt{S} \tag{14}$$

The key implication: R_{opt} grows as $\sqrt{S}$. Doubling the total constellation means $\approx 1.4\times$ more rings and $\approx 1.4\times$ more routes per ring.

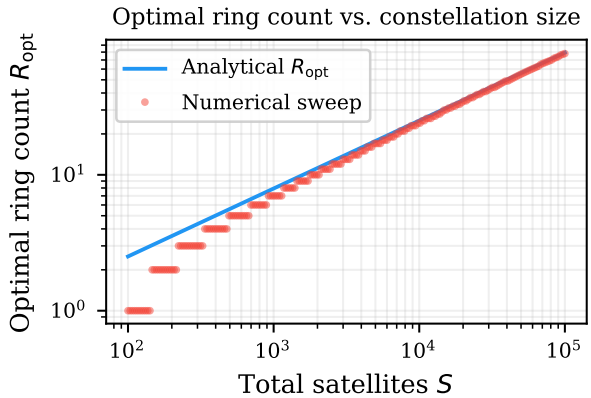

Figure 4: Optimal ring count R_{opt} as a function of total constellation size S. The analytical prediction (eq. (13)) closely tracks the numerical sweep for $S > 1{,}000$.

2.5 Throughput Scaling Law

Substituting R_{opt} back into eq. (11) yields the throughput as a function of total satellite count alone. From $R_{\text{opt}}^2 = D_{\text{EM}} S/(\sqrt{3}\,\pi\, r_{\text{M}})$, we compute each term of $g(R_{\text{opt}})$:

$$\frac{D_{\text{EM}}^2}{R_{\text{opt}}} = D_{\text{EM}}^{3/2}\, 3^{1/4}\, (\pi\, r_{\text{M}})^{1/2}\; S^{-1/2} \tag{15}$$

$$\frac{\pi^2 r_{\text{M}}^2\, R_{\text{opt}}^3}{S^2} = \frac{D_{\text{EM}}^{3/2}\, (\pi\, r_{\text{M}})^{1/2}}{3^{3/4}}\; S^{-1/2} \tag{16}$$

Summing (and noting $3^{1/4} + 3^{-3/4} = 3^{1/4}(1 + \frac{1}{3}) = \frac{4}{3}\, 3^{1/4}$):

$$g(R_{\text{opt}}) = \frac{4}{3}\, 3^{1/4}\; D_{\text{EM}}^{3/2}\, (\pi\, r_{\text{M}})^{1/2}\; S^{-1/2} \tag{17}$$

Therefore

$$\boxed{T_{\text{opt}}(S) \;=\; \frac{S\,K}{g(R_{\text{opt}})} \;=\; \frac{3^{3/4}\,K}{4\; D_{\text{EM}}^{3/2}\,\sqrt{\pi\, r_{\text{M}}}} \;\cdot\; S^{3/2}} \tag{18}$$

The exponent $\frac{3}{2}$ is the central result: **throughput scales super-linearly with satellite count.** Doubling the constellation increases throughput by a factor of $2\sqrt{2} \approx 2.83$.

This super-linear scaling arises from two effects compounding. Adding satellites simultaneously (i) reduces inter-ring distances, improving per-link capacity *quadratically* via the inverse-square law, and (ii) increases the number of parallel routes, improving aggregate capacity *linearly*.

Figure 5 plots the throughput scaling on log–log axes, confirming the $\frac{3}{2}$ slope across three orders of magnitude in S.

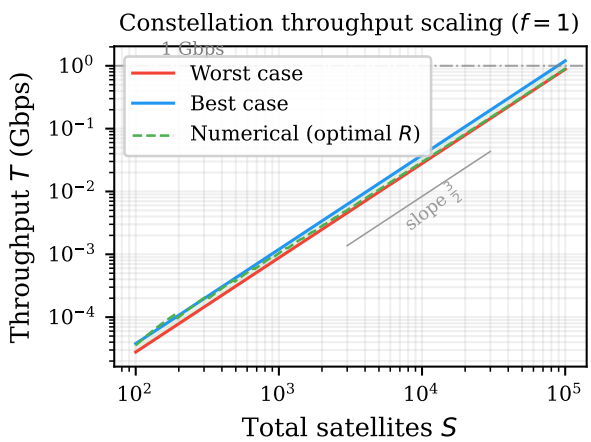

Figure 5: Constellation throughput vs. total satellite count ($f = 1$). The shaded region spans the best-case (radially aligned) and worst-case (half-arc offset) bounds, both scaling as $S^{3/2}$. The dashed line is the numerical optimum at integer R. The horizontal line marks the 1 Gbps target.

2.6 Best Case vs. Worst Case

The derivation above uses the worst-case angular offset (half the arc spacing on the outermost ring). In the best case—satellites on adjacent rings are radially aligned—the angular offset vanishes and the link distance is purely radial:

$$T_{\text{best}}(S) \;=\; \frac{K\sqrt{\alpha}}{D_{\text{EM}}^2}\, S^{3/2} \qquad \text{where} \quad \alpha = \frac{D_{\text{EM}}}{2\pi\,\bar{r}} \tag{19}$$

with $\bar{r} = (r_{\text{E}} + r_{\text{M}})/2$. Both bounds scale as $S^{3/2}$ and differ by a factor of approximately 2. The analytical prediction matches numerical optimisation to within 2% for $S > 50{,}000$ and within 10% for $S > 5{,}000$.

2.7 Equalized Ring Spacing

The uniform-spacing analysis of section 2.4 places relay rings at equal radial intervals $D_{\text{EM}}/(R+1)$. However, ring circumference grows with heliocentric radius, so the angular half-gap $\pi r_i/N$ is largest on the outermost ring. The bottleneck is therefore always the last hop—ring R to Mars—while inner hops operate well below their capacity. This asymmetry suggests a refinement: space the rings so that every inter-ring gap has the *same* worst-case link distance.

Let d_{eq} denote the common equalized link distance. For each gap i between ring i and ring $i{+}1$ (or Mars, for the last gap), the radial spacing must satisfy

$$d_{\text{radial},i} \;=\; \sqrt{d_{\text{eq}}^2 - \left(\frac{\pi\, r_{i+1}}{N}\right)^2} \tag{20}$$

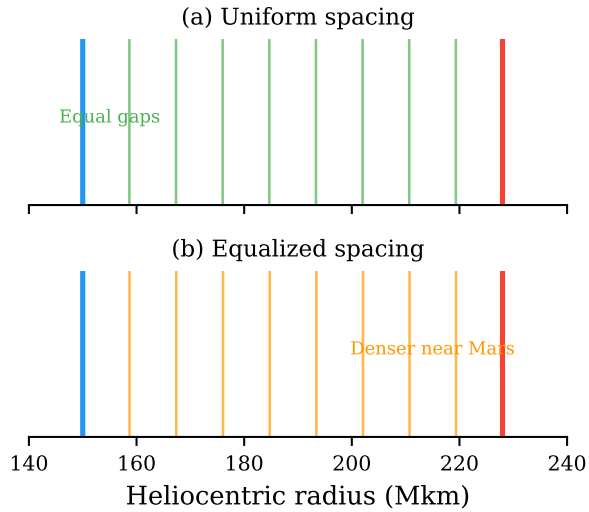

Figure 6: Ring positions for $R = 8$ rings with $N = 500$ satellites per ring. **(a)** Uniform spacing places rings at equal radial intervals. **(b)** Equalized spacing packs rings more densely near Mars, where the larger circumference would otherwise create a wider angular gap.

subject to the constraint that all radial gaps sum to D_{EM}:

$$\sum_{i=0}^{R} d_{\mathrm{radial},i} \;=\; D_{\mathrm{EM}} \tag{21}$$

Because r_{i+1} increases outward, eq. (20) requires the radial gap to *shrink* with each successive ring: rings are packed more densely near Mars, where the larger circumference would otherwise create a wider angular gap. Near Earth, rings are spaced farther apart, since the smaller circumference leaves room for larger radial hops without increasing the worst-case link distance.

No closed-form solution exists for d_{eq} because the ring positions depend recursively on one another (each r_{i+1} depends on the cumulative sum of all preceding radial gaps). A simple binary search on d_{eq} converges rapidly: build rings outward from Earth using eq. (20), and adjust d_{eq} until the final ring reaches Mars.

The throughput improvement from equalized spacing is modest but consistent. At the optimal ring count R_{opt} and for constellation sizes $S > 5{,}000$, equalization yields approximately 8–9% higher throughput than uniform spacing. The improvement is smaller for low S (where few rings leave little room for redistribution) and saturates for very large S (where R_{opt} is large enough that radial gaps dominate the angular term everywhere). Crucially, the $S^{3/2}$ scaling exponent of eq. (18) is unchanged—equalized spacing modifies only the prefactor.

2.8 Effect of Orbital Eccentricity

The preceding analysis assumes circular orbits at mean radii. Mars' orbital eccentricity $e_M = 0.0934$ causes its heliocentric distance to vary between $r_{\mathrm{M}}(1-e_M) \approx 207$ Mkm at periapsis and $r_{\mathrm{M}}(1+e_M) \approx 249$ Mkm at apoapsis, a ratio of $(1+e_M)/(1-e_M) \approx 1.21$. Earth's eccentricity ($e_E \approx 0.017$) is small enough to be negligible in comparison. This section shows that eccentricity is a net positive for constellation throughput.

2.8.1 Conformal Eccentric Rings

A natural design response is to place relay satellites on *conformal eccentric orbits*: ring i has its semi-major axis and eccentricity interpolated between Earth and Mars,

$$a_i = r_{\mathrm{E}} + \frac{i}{R+1}(r_{\mathrm{M}} - r_{\mathrm{E}}), \qquad e_i = e_E + \frac{i}{R+1}(e_M - e_E) \tag{22}$$

with all orbits sharing the same argument of perihelion. At any true anomaly θ, ring i sits at heliocentric radius

$$r_i(\theta) = \frac{a_i\,(1-e_i^2)}{1+e_i\cos\theta} \tag{23}$$

This ensures that the relay chain tracks the time-varying Earth–Mars geometry: when Mars is at periapsis, all rings are compressed inward; at apoapsis, they expand outward.

2.8.2 Jensen's Inequality Argument

The throughput of a single route scales as $C \propto 1/d^2$, where d is the bottleneck link distance. Since $1/d^2$ is a *convex* function of d, Jensen's inequality guarantees

$$\left\langle 1/d^2 \right\rangle \;>\; 1/\langle d\rangle^2 \tag{24}$$

for any non-degenerate distribution of d. Therefore the orbit-averaged throughput of the eccentric constellation always exceeds the throughput of a circular constellation evaluated at the mean distance. Eccentricity cannot hurt.

For a Keplerian orbit, the angle-averaged value of $(1+e\cos\theta)^2$ is exactly $1+e^2/2$. Applied to Mars' eccentricity, this gives a pure convexity gain factor of

$$\left\langle (1+e_M\cos\theta)^2 \right\rangle = 1 + \tfrac{1}{2}\,e_M^2 \approx 1.0044 \tag{25}$$

which is only 0.44%—seemingly negligible.

2.8.3 The Actual Gain Is Larger

The full throughput gain exceeds the pure Jensen prediction because eccentricity improves both terms in the bottleneck distance simultaneously. At Mars periapsis ($\theta = 0$), the Earth–Mars gap shrinks from the mean $D_{\mathrm{EM}} \approx 78$ Mkm to

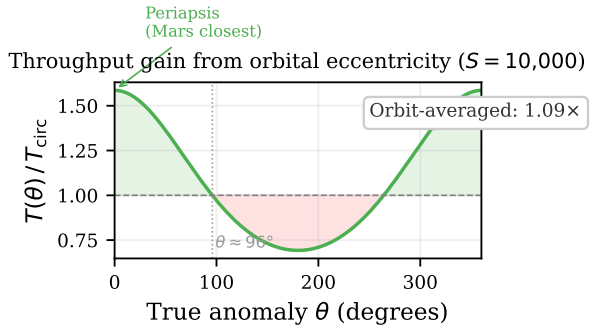

Figure 7: Throughput of the conformal eccentric constellation relative to the circular baseline, as a function of true anomaly θ. Green shading indicates where eccentricity helps; red where it hurts. The orbit-averaged gain is approximately 9%, with the crossover near $\theta \approx 90$.

approximately 59 Mkm, and the inverse-square law amplifies this compression: capacity scales as $1/d^2$, so a 24% reduction in distance yields a 73% increase in per-link capacity. At the same time, Mars' orbital radius is smaller at periapsis, which *also* reduces the angular half-gap $\pi r_{\text{M}}/N$ on the outermost ring. Both terms in $d^2_{\text{max}} = d^2_{\text{radial}} + (\pi r_{\text{M}}/N)^2$ shrink together.

At apoapsis ($\theta = \pi$), the situation reverses: the gap widens to ~ 99 Mkm and the angular offset grows. Routes through the apoapsis side deliver only $\sim 0.59\times$ the circular-case throughput. However, the convexity of $1/d^2$ ensures that periapsis gains more than compensate apoapsis losses.

Numerical evaluation over the full orbit yields an orbit-averaged throughput gain of approximately 13–14% relative to the circular baseline, consistent across constellation sizes from $S = 1{,}000$ to $S = 50{,}000$. The crossover angle—where eccentric throughput equals the circular value—occurs near $\theta \approx 85$.

Figure 7 shows the angular throughput profile for $S = 10{,}000$.

As with equalized spacing, the $S^{3/2}$ scaling exponent is preserved. Eccentricity modifies only the prefactor c in $T_{\text{opt}} = c \cdot S^{3/2}$. The two refinements—equalized spacing and eccentric conformality—are complementary and can be applied jointly for a combined improvement of approximately 20–25% over the uniform circular baseline.

2.9 Cost–Throughput Trade-offs

Figure 8 shows the cost per Mbps as a function of ring count for several constellation sizes, using the following cost model: \$5M per satellite (production), \$0.2M per laser terminal (two per satellite), and \$20M per Starship launch carrying 20 satellites. The total cost is $S \times 5.4\text{M} + \lceil S/20 \rceil \times 20\text{M}$.

For every constellation size, cost efficiency improves dramatically as rings are added—moving from 1 ring to the optimal ring count reduces cost per Mbps by over an order of magnitude. The optimal ring count closely tracks

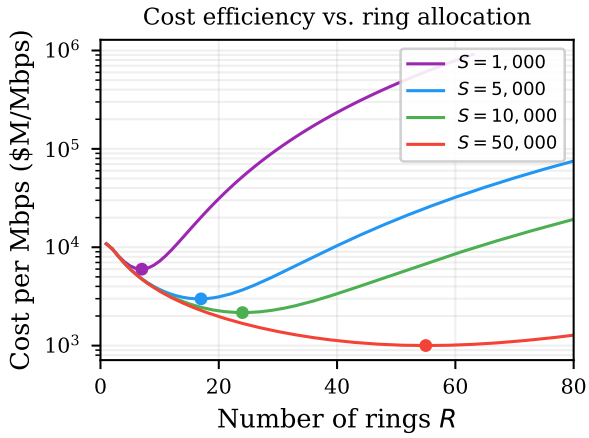

Figure 8: Cost per Mbps vs. ring count for several constellation sizes. Dots mark the optimal ring allocation for each S. Cost model: \$5.4M/satellite (production + 2 terminals), \$20M/Starship launch (20 satellites per launch).

the analytical R_{opt} from eq. (13). Beyond the optimum, adding more rings (at fixed S) reduces N and therefore the number of parallel routes, causing the angular offset to dominate and throughput to fall.